television screen when you disconnect the cable feed and tune to a station that has ceased its scheduled broadcasts is, in fact, due to this dim aftermath of the big bang. This match between theory and experiment confirms the big bang picture of cosmology as far back as the time that photons first moved freely through the universe, about a few hundred thousand years after the bang (ATB).

Can we push further in our tests of the big bang theory to even earlier times? We can. By using standard principles of nuclear theory and thermodynamics, physicists can make definite predictions about the relative abundance of the light elements produced during the period of primordial nucleosynthesis, between a hundredth of a second and a few minutes ATB. According to theory, for example, about 23 percent of the universe should be composed of helium. By measuring the helium abundance in stars and nebulae, astronomers have amassed impressive support that, indeed, this prediction is right on the mark. Perhaps even more impressive is the prediction and confirmation regarding deuterium abundance, since there is essentially no astrophysical process, other than the big bang, that can account for its small but definite presence throughout the cosmos. The confirmation of these abundances, and more recently that of lithium, is a sensitive test of our understanding of early universe physics back to the time of their primordial synthesis.

This is impressive almost to the point of hubris. All the data we possess confirm a theory of cosmology capable of describing the universe from about a hundredth of a second ATB to the present, some 15 billion years later. Nevertheless, one should not lose sight of the fact that the newborn universe evolved with phenomenal haste. Tiny fractions of a second—fractions much smaller than a hundredth of a second—form cosmic epochs during which long-lasting features of the world were first imprinted. And so, physicists have continued to push onward, trying to explain the universe at ever earlier times. Since the universe gets ever smaller, hotter, and denser as we push back, an accurate quantum-mechanical description of matter and the forces becomes increasingly important. As we have seen from other viewpoints in earlier chapters, point-particle quantum field theory works until typical particle energies are around the Planck energy. In a cosmological context, this occurred when the whole of the known